Split horoballs and the necessity direction of medial-axis reconstruction on Hadamard manifolds*

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Abstract

Conjecture 5.1 of [2] proposed that on a Hadamard manifold every reconstructible point of X^c lies in the horoball hull $\widehat{X}$ or in a *defective* maximal free horoball. The hull-based notion of defectiveness is inadequate: the horoball hull can collapse to X itself (already for a two-point set in $\mathbb{H}^2$), destroying all defectiveness while reconstruction survives. We replace defectiveness by tangency data on the horosphere — ultimately the *split* condition of §7 — and prove the structural half of the corresponding necessity statement. The key negative result (Proposition 3.1) is a rigidity statement: inflating a medial ball along the geodesic through one of its feet is cut *instantly* by any other foot, so escaping inflations exist only from uniquely-footed centres and produce horoballs with a single attained contact — bitangency can only be created by equidistant motion of centres. The positive results: a drift theorem (Theorem 4.2) — medial balls of unbounded radius containing p with uniform margin converge, via the horofunction boundary, to a maximal free horoball containing p at definite depth, whose attained contacts are the subsequential limits of feet — and an unconditional structure trichotomy (Theorem 5.1): *every* point of X^c lies in $\widehat{X}$, or satisfies the drift hypothesis, or lies at definite depth in single-contact escape horoballs; there is no residual case. In $\mathbb{R}^n$, reconstructible points outside the hull always drift (Proposition 5.3, by lifting the defect witness of [3]). The arguments use no entry points into $\widehat{X}$ and are immune to the tangential-entry gap of [2, §5]. We close with the taxonomy separating bitangency from defectiveness, an example (a horocyclic arc in $\mathbb{H}^2$) showing that bitangency alone does *not* imply reconstruction, the corrected notion of a *split* horoball (separated contact clusters), a complete proof that split horoballs are reconstructed whenever the splitting scale exceeds half the asymptotic separation of the contact rays — unconditionally under strictly negative curvature —, a deepest-direction dichotomy reducing the remaining necessity gap to an asymptotic-tangency regime along the deepest axis, its resolution on $\mathbb{H}^2$ for non-degenerate drift, an asymptotic-strand example showing that defectiveness survives exactly where splitness fails. Finally (§9) we show that the split-or-defective conjecture is *false* for $n \geq 3$: in $\mathbb{H}^3$ the unit circle on a horosphere has collapsing hull and only connected (hence non-split, non-defective) contact sets, yet $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0 \neq \varnothing$. Splitness is a two-dimensional artefact: the operative condition is failure of the contact set to be convex inside the intrinsically flat horosphere $\cong \mathbb{E}^{n-1}$, which on a line ($n = 2$) means a gap (split) but for $n \geq 3$ is implied already by a connected set such as a circle. We propose the corrected notion — horospherical defect — and prove its sufficiency direction in $\mathbb{H}^n$ via a duality with Motzkin's theorem (a closed Euclidean set is convex iff Chebyshev): every horospherically defective horoball with

*Companion note to [2], which generalised the sufficiency half of Theorem 6 of [1] to Riemannian manifolds, and to our formalization notes [3] on the Euclidean case. Drafted with the assistance of Claude (Anthropic); all arguments were re-derived and checked by hand. Remaining errors are ours.

attained, bulk-isolated contacts satisfies $\mathcal{H} \subseteq \mathcal{R}_X$, and when X lies on the horosphere this is sharp ($\mathcal{R}_X = \varnothing$ for convex X). In particular the circle counterexample is fully resolved — $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0$ is exactly the union of its horospherically defective horoballs — so the example that refuted the old conjecture confirms its repair. For necessity we prove the non-merging estimate: the two feet of a reconstructing drift, once they ascend to the horosphere (Lemma 11.2), either collapse onto a single contact or stay separated by exactly $\operatorname{arccosh}(1 + \frac{1}{2}|\kappa - \kappa'|^2)$, with no intermediate rate; in the non-collapse case their distinct limits are equidistant-nearest contacts, so by Motzkin K is non-convex and the horoball is horospherically defective (Theorem 11.3). This proves necessity in $\mathbb{H}^n$ for every bounded, non-collapsing drift, reducing the whole conjecture to a single residual asymptotic/local-hull question. A status table closes the note.

1 Setting and definitions

Throughout, (M^n, g) is a *Hadamard manifold*: complete, simply connected, of nonpositive sectional curvature. We write $d(y, z)$ for the Riemannian distance. Every geodesic is globally minimizing, extends to a complete geodesic line, and is determined by a point and a velocity. $X \subsetneq M$ is a fixed nonempty closed set, $d(y) := \operatorname{dist}(y, X)$, and

$$m(y) := \{x \in X \mid d(y, x) = d(y)\}$$

is the nonempty compact set of *feet* of y (Hopf–Rinow). All balls $B(y, r)$ are open. A ball (or any set) is *free* if it is disjoint from X; thus $B(y, r)$ is free iff $r \leq d(y)$.

Definition 1.1 (Medial axis, reconstructible set)**.** *The* medial axis *is* $M_X := \{a \in X^c : \#m(a) \geq 2\}$*; on a Hadamard manifold this coincides with the set of points admitting at least two distinct minimizing geodesics to* X *(no branching, cf. [2, Rem. 2.2]). The* reconstructible set *is*

$$\mathcal{R}_X := \bigcup_{a \in M_X} B\big(a, d(a)\big).$$

For $p \in \mathcal{R}_X$ *and* $a \in M_X$ *with* $p \in B(a, d(a))$ *we call* $\mu(a, p) := d(a) - d(p, a) > 0$ *the* margin *of* p *in the medial ball* $B(a, d(a))$*.*

Definition 1.2 (Busemann function, maximal free horoball)**.** *For a unit-speed geodesic ray* $\gamma : [0, \infty) \to M$ *the* Busemann function *is* $b_\gamma(y) := \lim_{t\to\infty}\big(d(y, \gamma(t)) - t\big)$*; the limit exists since the expression is non-increasing in* t *and bounded below by* $-d(y, \gamma(0))$*. It is* 1*-Lipschitz, convex, and* $b_\gamma(\gamma(s)) = -s$*. Set* $c_\gamma := \inf_X b_\gamma$*. When* c_γ *is finite, the* maximal free horoball *and its* horosphere *are*

$$\mathcal{H}_\gamma := \{b_\gamma < c_\gamma\}, \qquad S_\gamma := \{b_\gamma = c_\gamma\}.$$

$\mathcal{H}_\gamma$ *is open, convex, free, and maximal among free sub-level sets of* b_γ*. The* depth *of a point* $p \in \mathcal{H}_\gamma$ *is* $c_\gamma - b_\gamma(p) > 0$*. The* horoball hull *is* $\widehat{X} := M \setminus \bigcup\{$*free open horoballs*$\} = \{y : b_\gamma(y) \geq c_\gamma$ *for every ray* $\gamma\}$*, a closed set containing* X*.*

Definition 1.3 (Bitangent; defective)**.** *A maximal free horoball* $\mathcal{H}_\gamma$ *is*

(a) bitangent *if* $S_\gamma \cap X$ *contains at least two points (each contact point is the foot of a distinct minimizing geodesic from any centre* $\gamma(t)$ *deep enough, so bitangency witnesses genuine multiplicity on Hadamard manifolds);*

(b) defective *(the notion of [2, Def. 2.5]) if* $(S_\gamma \cap \widehat{X}) \setminus X \neq \emptyset$.

Remark 1.4 (Why defectiveness must be replaced). For $X = \{x_1, x_2\} \subset \mathbb{H}^2$ one has $\widehat{X} = X$: in the upper half-plane model with $x_{1,2} = (\pm s, 1)$, every point $z = (u, v) \notin X$ with $v \le 1$ lies in the free horoball tangent to the ideal boundary at $(u, 0)$ of Euclidean radius $\rho = \frac{1}{2}\big(\min_\pm (u \mp s)^2 + 1\big)$, and every point with $v > 1$ lies in the free horoball $\{y > 1\}$. (This corrects Remark 2.6 of [2]: the horoball hull does not even contain the geodesic segment $[x_1, x_2]$, whose interior dips into $\{y > 1\}$.) Consequently no maximal free horoball is defective, although $\mathcal{R}_X = \{y > 1\} \cup B_0 \neq \emptyset$, where B_0 is the horoball tangent at 0 through both points: Conjecture 5.1 of [2] fails as stated. Both horoballs constituting $\mathcal{R}_X$ are bitangent, which motivates Definition 1.3(a) and the revision below. We do not reproduce the full computation here, the original conjecture having been a working formulation.

Conjecture 1.5 (Revised necessity-and-sufficiency)**.** *Let M be Hadamard and X closed. Then*

$$\mathcal{R}_X \;=\; (\widehat{X} \setminus X) \;\cup\; \bigcup \Big\{ \mathcal{H}_\gamma \;:\; \mathcal{H}_\gamma \text{ maximal free, split (Def. 7.2) or defective} \Big\}.$$

The inclusion $\supseteq$ holds for $\widehat{X} \setminus X$ by [2, Thm. 3.1] (completeness only) and for defective horoballs by [2, Thm. 4.1]; for split horoballs it is proved under a separation-scale condition in Theorem 7.5 below. Bitangency alone is *not* sufficient (Example 7.1), which is what forces the split refinement. The present note proves the structural half of $\subseteq$ (Proposition 3.1, Theorems 4.2 and 5.1) and the split half of $\supseteq$ (Theorem 7.5). One external input is used, as in [2]:

Lemma 1.6 (Cut-locus density; Wolter, Albano [6, 5])**.** *Let M be complete and X closed. The cut locus* $\mathrm{Cut}(X)$ *— the set of endpoints of maximal minimizing segments of normal geodesics from X — satisfies* $\mathrm{Cut}(X) \subseteq \overline{M_X}$.

2 The inflation dichotomy, with margins

We first record the inflation construction of [2, Thm. 3.1] in the quantitative form we need. Fix $a \in X^c$, a foot $q \in m(a)$, and let $\gamma : [0, \infty) \to M$ be the unit-speed ray with $\gamma(0) = q$ and $\gamma(\rho) = a$, where $\rho := d(a)$ (the geodesic from q through a, extended; on a Hadamard manifold the extension is a globally minimizing ray). Define the set of *free times* $T := \{t > 0 : B(\gamma(t), t) \cap X = \emptyset\}$ and the *escape time*

$$r(a, q) := \sup T \;\in\; [\rho, \infty].$$

Lemma 2.1 (Monotone free inflation)**.** *T is an interval containing $(0, \rho]$, and for $t' < t$ with $t \in T$ one has $B(\gamma(t'), t') \subseteq B(\gamma(t), t)$. Moreover for every $t \in T$:*

(i) $d(\gamma(t)) = t$*, with* $q \in m(\gamma(t))$*;*

(ii) if $p \in B(a, \rho)$ with margin $\mu = \rho - d(p, a)$, then $d(p, \gamma(t)) \le t - \mu$ for all $t \ge \rho$, $t \in T$;

(iii) every $q' \in m(a)$ satisfies $d(q', \gamma(t)) = t$; in particular all feet of a lie on $\partial B(\gamma(t), t)$.

Proof. Since $d(\gamma(t'), \gamma(t)) = t - t'$ (geodesics minimize), for $y \in B(\gamma(t'), t')$ we get $d(y, \gamma(t)) \le d(y, \gamma(t')) + (t - t') < t$, proving the nesting and that T is an interval; $\rho \in T$ because $B(a, \rho)$ is free. (i): freeness gives $d(\gamma(t)) \ge t$, while $d(\gamma(t), q) = t$ gives $\le$ and exhibits q as a foot. (ii): $d(p, \gamma(t)) \le d(p, a) + d(a, \gamma(t)) = (\rho - \mu) + (t - \rho)$. (iii): $d(q', \gamma(t)) \le d(q', a) + (t - \rho) = \rho + (t - \rho) = t$ by the triangle inequality, and $\ge t$ by freeness, since $q' \in X$. $\square$

Part (iii) looks like an engine for producing bitangent limit horoballs; in fact it is self-limiting. Boundary-riding is an *equality* in the triangle inequality, and equalities in uniquely geodesic spaces are rigid: they force collinearity, which destroys freeness immediately. Section 3 turns this into a rigidity statement that any necessity mechanism must respect.

Lemma 2.2 (Cut branch)**.** *Suppose $r := r(a,q) < \infty$. Then $c := \gamma(r)$ satisfies $d(c) = r$, the ball $B(c,r)$ is free and contains $B(a,\rho)$, and $c \in \mathrm{Cut}(X) \subseteq \overline{M_X}$. Consequently, if $p \in B(a,\rho)$ has margin μ, then for every $\varepsilon \in (0,\mu/2)$ there exists $a' \in M_X$ with*

$$d(a') \geq r - \varepsilon, \qquad p \in B\big(a', d(a')\big), \qquad \mu(a',p) \geq \mu - 2\varepsilon.$$

Proof. $d(\gamma(t)) = t$ on T and continuity of d give $d(c) = r$. If $y \in B(c,r)$, pick $\eta > 0$ with $d(y,c) < r - \eta$; for $t < r$ with $r - t < \eta/2$, $d(y,\gamma(t)) \leq d(y,c) + (r-t) < r - \eta + \eta/2 < t$, so $y \in B(\gamma(t),t)$, which is free; hence $B(c,r)$ is free, and it contains each $B(\gamma(t),t) \supseteq B(a,\rho)$ by Lemma 2.1. The segment $\gamma|_{[0,t]}$ minimizes $d(\cdot,X)$ for every $t \leq r$ by Lemma 2.1(i), while for $t > r$ freeness fails, so $d(\gamma(t)) < t$ and the segment no longer minimizes: c is the cut point of X along γ, and Lemma 1.6 applies. Finally take $a' \in M_X$ with $d(a',c) < \varepsilon$: then $d(a') \geq d(c) - \varepsilon = r - \varepsilon$ and $d(p,a') \leq d(p,c) + \varepsilon \leq (r-\mu) + \varepsilon$ using Lemma 2.1(ii) at $t \to r$; hence $\mu(a',p) = d(a') - d(p,a') \geq \mu - 2\varepsilon > 0$. ☐

3 Rigidity: no escape from a medial centre

Proposition 3.1 (Instant cutting)**.** *Let M be Hadamard and $a \in M_X$. Then $r(a,q) = d(a)$ for every foot $q \in m(a)$: the inflation of a medial ball along any of its foot-geodesics is cut instantly.*

Proof. Write $\rho := d(a)$, pick a second foot $q' \neq q$, and suppose $t \in T$ with $t > \rho$. By Lemma 2.1(iii), $d(q',\gamma(t)) = t = d(q',a) + d(a,\gamma(t))$: equality in the triangle inequality, so the concatenation of the geodesic segments $[q',a]$ and $[a,\gamma(t)]$ has length $d(q',\gamma(t))$ and is therefore the (unique) geodesic from q' to $\gamma(t)$. In particular it passes through a with velocity $\dot\gamma(\rho)$, so its initial segment $[q',a]$, reversed, is part of γ: $q' = \gamma(\rho - d(q',a)) = \gamma(0) = q$, contradicting $q' \neq q$. Hence $T \subseteq (0,\rho]$ and $r(a,q) = \rho$. ☐

Remark 3.2. Proposition 3.1 redirects the search for a necessity mechanism. Escaping inflations cannot start at medial centres, so a single inflation can never tag two feet on a limit horosphere; bitangency must instead be produced by moving centres *along equidistant loci* of two contact clusters — which is exactly what the pincer in the proof of Theorem 7.5 does, and what the medial families of the worked examples (e.g. the bisector family of a two-point set in $\mathbb{H}^2$) actually look like. An earlier draft of this note stated a bitangency theorem for escaping inflations from medial centres; the rigidity above shows its hypothesis is unsatisfiable.

What survives of the escape mechanism is its single-contact version, which by Proposition 3.1 can only fire at uniquely-footed centres; it is used in the proof of the structure theorem below.

Theorem 3.3 (Single-foot escape)**.** *Let M be Hadamard, X closed, $a \in X^c$, $q \in m(a)$, and let $p \in B(a,d(a))$ have margin $\mu > 0$. If $r(a,q) = \infty$ — which forces $m(a) = \{q\}$ — then with γ the ray of §2:*

(a) $c_\gamma = 0$, attained at q: $\mathcal{H}_\gamma = \{b_\gamma < 0\}$ is a maximal free horoball with attained contact $q \in S_\gamma \cap X$;

(b) $b_\gamma(p) \le -\mu$*, so* $p \in \mathcal{H}_\gamma$ *at depth at least* μ*;*

(c) $\mathcal{H}_\gamma = \bigcup_{t>0} B(\gamma(t), t)$*, an increasing union of free balls each containing* $B(a, d(a))$ *for* $t \ge d(a)$*.*

Proof. All balls $B(\gamma(t), t)$, $t > 0$, are free. For $x \in X$ this means $d(x, \gamma(t)) \ge t$, hence $b_\gamma(x) \ge 0$ and $c_\gamma \ge 0$. By Lemma 2.1(i), $d(q, \gamma(t)) = t$ for all t, so $b_\gamma(q) = 0$; with $q \in X$ this forces $c_\gamma = 0$, attained at q, proving (a). By Lemma 2.1(ii), $d(p, \gamma(t)) \le t - \mu$ for all $t \ge d(a)$, whence $b_\gamma(p) \le -\mu < 0 = c_\gamma$, proving (b). For (c), the inclusion $\subseteq$ of the union follows from $b_\gamma(y) \le d(y, \gamma(t)) - t < 0$ for $y \in B(\gamma(t), t)$; conversely if $b_\gamma(y) < 0$ then $d(y, \gamma(t)) < t$ for all large t. The containment statement is Lemma 2.1. □

4 Branch II: unbounded medial radii

By Proposition 3.1, escaping inflations cannot start from medial centres; the mechanism by which the medial axis actually produces horoballs is *drift*: medial balls containing p with radii tending to infinity and uniform margin (realized, for instance, by the bisector family of a two-point set in $\mathbb{H}^2$). For this branch we use the horofunction description of the boundary at infinity. For $\delta > 0$ set

$$T_\delta(p) := \sup\{\, d(a) : a \in M_X,\ \mu(a, p) \ge \delta \,\} \in [0, \infty], \qquad \sup \emptyset := 0.$$

Lemma 4.1 (Horofunction convergence; [4, II.8.13])**.** *Let* M *be Hadamard,* $o \in M$*, and let* $(a_k) \subset M$ *with* $d(o, a_k) \to \infty$ *and* $a_k \to \xi \in \partial_\infty M$ *in the cone topology. Then* $h_k(\cdot) := d(\cdot, a_k) - d(o, a_k) \longrightarrow b_\xi(\cdot)$ *uniformly on compact sets, where* b_ξ *is the Busemann function of the ray from* o *to* ξ*, normalized by* $b_\xi(o) = 0$*. Moreover* $\overline{M} = M \cup \partial_\infty M$ *is compact, so every unbounded sequence subconverges to some* $\xi \in \partial_\infty M$*.*

Theorem 4.2 (Limit horoball from unbounded medial radii)**.** *Let* M *be Hadamard,* X *closed,* $p \in X^c$*, and suppose* $T_\delta(p) = \infty$ *for some* $\delta > 0$*. Then there exist* $\xi \in \partial_\infty M$*,* $s_\infty \in [\delta, d(p)]$*, medial centres* $a_k \in M_X$ *with* $r_k := d(a_k) \to \infty$ *and* $\mu(a_k, p) \ge \delta$*, such that, with* b_ξ *normalized at* $o = p$*:*

(a) $X \subseteq \{b_\xi \ge s_\infty\}$ *and* $b_\xi(p) = 0 \le s_\infty - \delta$*; hence* $c_\xi \ge s_\infty > 0 = b_\xi(p)$ *and* p *lies in the maximal free horoball* $\mathcal{H}_\xi = \{b_\xi < c_\xi\}$ *at depth at least* δ*;*

(b) the free balls exhaust the sub-horoball: every compact $K \subseteq \{b_\xi < s_\infty\}$ *satisfies* $K \subseteq B(a_k, r_k)$ *for all large* k*;*

(c) (contact capture) *if feet* $q_k \in m(a_k)$ *have a bounded subsequence, then along it* $q_k \to q_* \in X$ *with* $b_\xi(q_*) = s_\infty$*; in that case* $c_\xi = s_\infty$*, the infimum is attained at* q_**, and* $q_* \in S_\xi \cap X$*. If two sequences of feet* $q_k \ne q'_k$ *of* a_k *subconverge to* distinct *limits,* $\mathcal{H}_\xi$ *is bitangent.*

Proof. Choose $a_k \in M_X$ with $\mu(a_k, p) \ge \delta$ and $r_k = d(a_k) \to \infty$. Set $t_k := d(p, a_k) \le r_k - \delta$ and $s_k := r_k - t_k \ge \delta$. For any $x_0 \in X$, freeness of $B(a_k, r_k)$ gives $d(x_0, a_k) \ge r_k$, so $d(x_0, p) \ge d(x_0, a_k) - t_k \ge s_k$; hence $s_k \le d(p)$, and after passing to a subsequence $s_k \to s_\infty \in [\delta, d(p)]$. Also $t_k \ge r_k - d(p) \to \infty$, so (a_k) is unbounded and, after a further subsequence, $a_k \to \xi \in \partial_\infty M$ (Lemma 4.1).

(a) For $x \in X$: $h_k(x) = d(x, a_k) - t_k \ge r_k - t_k = s_k$, and letting $k \to \infty$, $b_\xi(x) \ge s_\infty$. Trivially $b_\xi(p) = 0$. Hence $c_\xi = \inf_X b_\xi \ge s_\infty \ge \delta$, while $c_\xi \le b_\xi(x_0) < \infty$; the maximal free horoball $\mathcal{H}_\xi$ exists and contains p at depth $c_\xi - b_\xi(p) \ge \delta$.

(b) If $b_\xi(y) < s_\infty$ then, by locally uniform convergence on a compact neighbourhood of K, $h_k(y) < s_k$ for large k uniformly on K, i.e. $d(y, a_k) < t_k + s_k = r_k$.

(c) Along the bounded subsequence, $h_k(q_k) = d(q_k, a_k) - t_k = r_k - t_k = s_k$, and locally uniform convergence plus $q_k \to q_*$ gives $b_\xi(q_*) = s_\infty$. Since $q_* \in X$ (closedness) we get $c_\xi \le s_\infty$, so $c_\xi = s_\infty$ is attained at q_*. Two distinct such limits give two points of $S_\xi \cap X$. □

Remark 4.3. Without the boundedness hypothesis in (c) the feet may escape to infinity and the contact be only asymptotic ($c_\xi = s_\infty$ not attained, or attained at fewer points than the medial multiplicity suggests). This is a genuine phenomenon already in $\mathbb{R}^2$: for $X = \operatorname{graph}(e^x) \cup \{(5,0)\}$ the medial centres $(\bar{x}_T, -R_T)$ with feet $(5,0)$ and $(-T, e^{-T})$ have $R_T \to \infty$, one foot converging to the attained contact $(5,0)$ and the other escaping along the asymptote; the limit half-plane $\{y < 0\}$ is maximal free with a single attained contact. See §6.

5 The structure theorem

Theorem 5.1 (Structure trichotomy — no residual case)**.** *Let M be Hadamard, X closed, and $p \in X^c$ (reconstructible or not). Then at least one of the following holds.*

- *(I)* ***(Hull)*** $p \in \widehat{X} \setminus X$.
- *(II)* ***(Drift)*** *$T_\delta(p) = \infty$ for some $\delta > 0$; then p lies, at depth $\ge \delta$, in a maximal free horoball exhausted (below level s_∞) by medial balls, whose attained contacts are the subsequential limits of their feet (Theorem 4.2).*
- *(III)* ***(Ray escape)*** *There is $\delta' > 0$ such that for all sufficiently large t, p lies at depth $\ge \delta'$ in a maximal free horoball with an attained contact, produced by an escaping single-foot inflation (Theorem 3.3).*

Proof. Assume (I) fails; since $p \notin X$, this means $p \notin \widehat{X}$. Choose a ray γ with $b_\gamma(p) < c_\gamma =: c$ (finite by this very inequality) and set $\delta^* := c - b_\gamma(p) > 0$. For $x \in X$ the function $s \mapsto d(x, \gamma(s)) - s$ is non-increasing with limit $b_\gamma(x) \ge c$, so $d(x, \gamma(t)) \ge t + c$ for every t: the balls $B(\gamma(t), t + c)$ are free, and $d(\gamma(t)) \ge t + c$. Since $d(p, \gamma(t)) - t \downarrow b_\gamma(p) = c - \delta^*$, there is t_0 with $d(p, \gamma(t)) \le t + c - \delta^*/2$ for $t \ge t_0$: p lies in the maximal free ball $B(\gamma(t), d(\gamma(t)))$ with margin $\mu_t \ge \delta^*/2$. For $t \ge t_0$ pick a foot $q_t \in m(\gamma(t))$. If $r(\gamma(t), q_t) < \infty$ for an unbounded set of t, Lemma 2.2 with $\varepsilon = \delta^*/8$ supplies $a'_t \in M_X$ with $d(a'_t) \ge d(\gamma(t)) - \delta^*/8 \ge t + c - \delta^*/8$ and $\mu(a'_t, p) \ge \delta^*/4$, so $T_{\delta^*/4}(p) = \infty$: case (II). Otherwise $r(\gamma(t), q_t) = \infty$ for all large t, and Theorem 3.3 (applied to $a = \gamma(t)$, margin $\delta^*/2$) gives case (III) with $\delta' = \delta^*/2$. □

Remark 5.2. The trichotomy is unconditional: it removes the "confinement" case conjectured in an earlier draft, and it does not require $p \in \mathcal{R}_X$. Reconstructibility enters only through the open *split upgrade* (Question 12.1): for $p \in \mathcal{R}_X$, the horoballs delivered by (II) and (III) must be improved to split-or-defective ones. Thus Conjecture 1.5 reduces to the split upgrade together with the sufficiency results of §7.

In Euclidean space the drift branch always carries reconstruction outside the hull:

Proposition 5.3 (In $\mathbb{R}^n$, reconstruction outside the hull forces drift)**.** *For $M = \mathbb{R}^n$ and $p \in \mathcal{R}_X \setminus \overline{\operatorname{conv}}\, X$, case (II) of Theorem 5.1 holds: $T_\delta(p) = \infty$ for some $\delta > 0$.*

Proof. By the repaired necessity direction of Theorem 6 [3, §4.6] there are a unit vector v, $c = \sup_X \langle v, \cdot\rangle$, and a defect witness $w \in (\overline{\operatorname{conv}}\, X \cap \{\langle v, \cdot\rangle = c\}) \setminus X$ with $h := \langle v, p\rangle - c > 0$. Lift the *witness*: $w_t := w + tv$. Then $d(w_t) \geq \langle v, w_t\rangle - c = t$ since $X \subseteq \{\langle v, \cdot\rangle \leq c\}$, while $|p - w_t|^2 = t^2 - 2th + |p - w|^2$ gives $|p - w_t| \leq t - h + O(1/t)$: for all large t, p lies in $B(w_t, d(w_t))$ with margin $\mu_t \geq h/2$.

No inflation from w_t escapes. Indeed, suppose $r(w_t, q_t) = \infty$ for a foot $q_t \in m(w_t)$, and let u_t be the direction of the inflation ray, so $q_t = w_t - \rho_t u_t$ with $\rho_t := d(w_t)$. Freeness of all balls $B(\gamma_t(s), s)$ gives, for every $x \in X$, $\langle x - q_t, u_t\rangle \leq 0$; the same then holds for every point of $\overline{\operatorname{conv}}\, X$, in particular for w. Substituting $w - q_t = \rho_t u_t - tv$:

$$0 \;\geq\; \langle w - q_t, u_t\rangle \;=\; \rho_t - t\,\langle v, u_t\rangle \;\geq\; \rho_t - t \;\geq\; 0,$$

forcing equality throughout: $\langle v, u_t\rangle = 1$, hence $u_t = v$, $\rho_t = t$, and $q_t = w_t - tv = w \notin X$ — contradicting $q_t \in X$.

Hence $r(w_t, q_t) < \infty$ for every t large, and Lemma 2.2 with $\varepsilon = h/8$ supplies $a'_t \in M_X$ with $d(a'_t) \geq d(w_t) - h/8 \geq t - h/8$ and $\mu(a'_t, p) \geq h/4$. Thus $T_{h/4}(p) = \infty$. □

Corollary 5.4 (Strengthened Euclidean necessity)**.** *For closed $X \subseteq \mathbb{R}^n$, every $p \in \mathcal{R}_X \setminus \overline{\operatorname{conv}}\, X$ lies at positive depth in a maximal free half-space L^+ that is a limit of medial balls in the sense of Theorem 4.2(b); the attained contacts of ∂L^+ are the subsequential limits of their feet. In particular the union in Theorem 6 of [1] may be taken over maximal free half-spaces arising as limits of medial balls.*

6 Bitangent versus defective

The two notions are incomparable in general; they agree for compact X in $\mathbb{R}^n$.

Proposition 6.1. *(a)* ***(Bitangent $\not\Rightarrow$ defective, Hadamard)*** *For X a two-point set in $\mathbb{H}^2$ (Remark 1.4), the two bitangent maximal free horoballs are not defective, since $\widehat{X} = X$.*

(b) ***(Defective $\not\Rightarrow$ bitangent, already in $\mathbb{R}^n$)*** *For $X = \operatorname{graph}(e^x) \cup \{(5,0)\} \subseteq \mathbb{R}^2$, the maximal free half-space $\{y < 0\}$ has the single attained contact $(5,0)$, yet it is defective: points $(3-\theta, 0)$, $\theta \geq 0$ small, lie in $\overline{\operatorname{conv}}\, X \setminus X$ on the boundary line. Mass escaping to infinity in the hull can even produce defective half-spaces whose witness is unreachable by contact sequences: for $X = \{(5,0)\} \cup \{(-k^2, -k) : k \in \mathbb{N}\}$, the half-space $\{y < -0\}$*[1] *is defective with witnesses $(5-c, 0)$, $c \in (0, C]$, obtained from convex combinations with weights $\sim 1/k$ on $(-k^2, -k)$, while every sequence in X with $y \to 0$ is eventually constant equal to $(5,0)$.*

(c) ***(Characterisation for compact X in $\mathbb{R}^n$)*** *If $X \subseteq \mathbb{R}^n$ is compact, a maximal free half-space $L^+ = \{\langle v, \cdot\rangle > c\}$, $c = \max_X \langle v, \cdot\rangle$, is defective if and only if the slice $\partial L^+ \cap X$ is* not convex. *In particular bitangency ($\#(\partial L^+ \cap X) \geq 2$) is necessary but not sufficient for defectiveness: a segment has a convex slice (cf. Example 7.1).*

Proof. (a) Remark 1.4. (b) The witness claims follow from computing limits of convex combinations as indicated; closedness of each X is immediate, and freeness and maximality of the half-spaces follow since inf is 0, attained at $(5,0)$ in both cases. (c) ($\Leftarrow$) If the slice is not convex,

[1] Precisely: the maximal free half-space in direction $-e_2$ through the contact $(5,0)$.

pick $x_1, x_2 \in \partial L^+ \cap X$ and $z \in [x_1, x_2] \setminus X$; then $z \in \operatorname{conv}(X \cap \partial L^+) \setminus X \subseteq (\overline{\operatorname{conv}}\, X \cap \partial L^+) \setminus X$ is a witness. ($\Rightarrow$) For compact X, $\operatorname{conv} X$ is compact and $\overline{\operatorname{conv}}\, X \cap \partial L^+ = \operatorname{conv}(X \cap \partial L^+)$: writing $w = \sum \lambda_i x_i$ with $x_i \in X$, $\langle v, w\rangle = c$ and $\langle v, x_i\rangle \le c$ force $\langle v, x_i\rangle = c$ whenever $\lambda_i > 0$. A defective witness $w \notin X$ then lies in $\operatorname{conv}(X \cap \partial L^+) \setminus (X \cap \partial L^+)$, so the slice differs from its convex hull: it is not convex. $\square$

Remark 6.2. Proposition 6.1(b) shows that Conjecture 1.5 genuinely needs the disjunction: defective horoballs cover reconstruction produced by asymptotic contacts (feet escaping to infinity, Remark 4.3), bitangent ones cover the hull-collapse regime where defectiveness is blind. Whether on a Hadamard manifold the drift branch with escaping feet always lands in a *defective* horoball is open; a robust common generalisation would count *contact ends* (limits of contact sequences in $M \cup \partial_\infty M$), but example (b) second part shows that even this undercounts the defective family in the presence of escaping deep mass. We leave the right asymptotic notion open (Question 12.2).

7 Sufficiency: split horoballs, and the failure of bare bitangency

Example 7.1 (Bitangency does not imply reconstruction). In the upper half-plane model of $\mathbb{H}^2$ let $X := \{(u, 1) : -1 \le u \le 1\}$, a compact horocyclic arc, and let γ be the upward vertical ray, so that $b_\gamma(x, y) = -\log y$ and $c_\gamma = 0$. The maximal free horoball $\mathcal{H}_\gamma = \{y > 1\}$ has contact set $S_\gamma \cap X = X$: it is bitangent with a continuum of contacts. Nevertheless $M_X = \emptyset$: for $(a, T) \notin X$,

$$\cosh d\big((a,T),(u,1)\big) \;=\; 1 + \frac{(u-a)^2 + (T-1)^2}{2T}$$

is strictly increasing in $(u - a)^2$, so the nearest point of the arc is the unique point with $u = \min(\max(a, -1), 1)$. Hence $\mathcal{R}_X = \emptyset \not\supseteq \mathcal{H}_\gamma$: *bitangent sufficiency is false*, and the second disjunct of Conjecture 1.5 must be the split family of Definition 7.2. The Euclidean shadow (X a segment in $\mathbb{R}^2$) is handled correctly by Theorem 6 of [1] because the boundary slice is convex and the half-space is not defective (Proposition 6.1(c)): what reconstruction detects is a *gap* in the contact set, not multiplicity.

Definition 7.2 (Split horoball). *A maximal free horoball $\mathcal{H}_\gamma$ is* split at scale $\eta > 0$ *if*

$$X \cap \{b_\gamma \le c_\gamma + \eta\} = F_1 \cup F_2, \qquad \operatorname{dist}(F_1, F_2) > 0, \qquad F_i \cap S_\gamma \ne \emptyset \;\; (i = 1, 2),$$

with F_1, F_2 closed and nonempty.

Remark 7.3. A split horoball is bitangent. In $\mathbb{R}^n$ a split half-space is defective: choose contacts $x_i \in F_i \cap \partial L^+$ and bisect repeatedly, always replacing the endpoint lying in the same cluster as the midpoint; the pair distance halves, so after finitely many steps it drops below $2\operatorname{dist}(F_1, F_2)$, at which point the midpoint can lie in neither cluster, hence not in X — a witness. Thus splitness is invisible in the Euclidean statement; on a Hadamard manifold it is the part of defectiveness that survives the collapse of the hull (Remark 1.4).

Lemma 7.4 (Pure descent from a contact). *Let $\mathcal{H}_\gamma$ be maximal free, normalised $c_\gamma = 0$, let $x_0 \in S_\gamma \cap X$, and let σ_0 be the asymptotic ray from x_0 to $\xi = \gamma(\infty)$. Then for all $t > 0$: $b_\gamma(\sigma_0(t)) = -t$, $d(\sigma_0(t)) = t$, and $m(\sigma_0(t)) = \{x_0\}$.*

Proof. By (H3), $b_\gamma \circ \sigma_0$ has slope -1 and vanishes at $t = 0$. For $x \in X$ the 1-Lipschitz bound gives $d(x, \sigma_0(t)) \geq b_\gamma(x) - b_\gamma(\sigma_0(t)) \geq t$, while $d(x_0, \sigma_0(t)) = t$; hence $d(\sigma_0(t)) = t$ with x_0 a foot. If x is any foot, equality $d(x, \sigma_0(t)) = t$ forces $b_\gamma(x) = 0$ and unit-rate descent of b_γ along the segment from x to $\sigma_0(t)$; with $|\nabla b_\gamma| \equiv 1$ (H1), the unit-speed segment satisfies $\dot{c} = -\nabla b_\gamma(c)$, so it is an integral curve of the locally Lipschitz field $-\nabla b_\gamma$ through $\sigma_0(t)$, as is σ_0; uniqueness of integral curves gives $x = \sigma_0(0) = x_0$ (cf. Step 1 of [2, Thm. 4.1]). □

Theorem 7.5 (Split sufficiency)**.** *Let M be Hadamard, X closed, and $\mathcal{H}_\gamma$ split at scale η with clusters F_1, F_2. Choose contacts $x_i \in F_i \cap S_\gamma$, let σ_i be the asymptotic rays from x_i to $\xi = \gamma(\infty)$, and set $L_\infty := \lim_{t\to\infty} d(\sigma_1(t), \sigma_2(t))$. If $\eta > L_\infty/2$, then $\mathcal{H}_\gamma \subseteq \mathcal{R}_X$.*

Proof. Normalise $c_\gamma = 0$ and write $b := b_\gamma$; by (H3), normalised at the contacts, $b_{\sigma_1} = b_{\sigma_2} = b$. The function $f(t) := d(\sigma_1(t), \sigma_2(t))$ is convex and bounded (the rays are asymptotic to the same ideal point), hence non-increasing, with $f(t) \downarrow L_\infty$. Fix t with $f(t) \leq 2\eta$ and let $J_t := [\sigma_1(t), \sigma_2(t)]$ be the geodesic segment.

Step 1: all feet over J_t lie in $F_1 \cup F_2$. Let $z \in J_t$ and $x \in m(z)$. Convexity of b along J_t gives $b(z) \leq \max_i b(\sigma_i(t)) = -t$. Since z lies on a segment of length $f(t)$, there is an i with $d(z, \sigma_i(t)) \leq f(t)/2$, whence $d(z) \leq d(z, x_i) \leq d(z, \sigma_i(t)) + d(\sigma_i(t), x_i) \leq t + f(t)/2$. The Lipschitz bound then yields $b(x) \leq b(z) + d(z, x) = b(z) + d(z) \leq f(t)/2 \leq \eta$, so $x \in X \cap \{b \leq \eta\} = F_1 \cup F_2$.

Step 2: a medial point $z_t \in J_t$ with $d(z_t) \geq t$. Set $A_i := \{z \in J_t : m(z) \subseteq F_i\}$. By Lemma 7.4, $\sigma_i(t) \in A_i$, so both sets are nonempty; they are disjoint because feet sets are nonempty and $F_1 \cap F_2 = \emptyset$. Each A_i is relatively open: if $z_k \to z$ in J_t and each z_k has a foot $x_k \in m(z_k) \setminus F_1$, then $x_k \in F_2$ by Step 1; the x_k are bounded ($d(z_k) \leq t + f(t)/2$ with z_k in the compact J_t), so a subsequence converges to some x with $d(z, x) = \lim d(z_k, x_k) = \lim d(z_k) = d(z)$ and $x \in F_2 \cap X$ (both closed); thus $z \notin A_1$, and contrapositively A_1 is relatively open; symmetrically A_2. Since J_t is connected, it is not the union of the two disjoint nonempty relatively open sets A_1, A_2: there is $z_t \in J_t$ with feet in both clusters, hence $\#m(z_t) \geq 2$ and $z_t \in M_X$. Finally, for every $x \in X$, $d(x, z_t) \geq b(x) - b(z_t) \geq t$, so $d(z_t) \geq t$.

Step 3: the medial balls $B(z_t, d(z_t))$ exhaust $\mathcal{H}_\gamma$. Let $p \in \mathcal{H}_\gamma$ and $\beta := -b(p) > 0$. The quantity $d(p, \sigma_i(t)) - t$ is non-increasing in t with limit $b_{\sigma_i}(p) = -\beta$, so $d(p, \sigma_i(t)) \leq t - \beta + \varepsilon(t)$ with $\varepsilon(t) \downarrow 0$. Convexity of $d(p, \cdot)$ along J_t gives $d(p, z_t) \leq \max_i d(p, \sigma_i(t)) \leq t - \beta + \varepsilon(t)$. For t large enough that $\varepsilon(t) < \beta$ and $f(t) \leq 2\eta$,

$$d(p, z_t) \ < \ t \ \leq \ d(z_t),$$

so $p \in B(z_t, d(z_t)) \subseteq \mathcal{R}_X$. □

Corollary 7.6. *If the contact rays satisfy $L_\infty = 0$ — in particular whenever M has sectional curvature $\leq -\kappa^2 < 0$, since asymptotic geodesics then converge — every split horoball, at every scale $\eta > 0$, is contained in $\mathcal{R}_X$.*

Beyond split horoballs, the following triviality records the sufficiency of the limit objects produced by the necessity branches:

Proposition 7.7. *If a maximal free horoball $\mathcal{H}$ is exhausted by medial balls (for every compact $K \subseteq \mathcal{H}$ there is $a \in M_X$ with $K \subseteq B(a, d(a))$), then $\mathcal{H} \subseteq \mathcal{R}_X$. In particular the drift horoballs of Theorem 4.2 satisfy $\{b < s_\infty\} \subseteq \mathcal{R}_X$: their approximating centres are medial by hypothesis. For the single-foot escape horoballs of Theorem 3.3 no such conclusion is available — consistently with their single attained contact and with Example 7.1.*

Proof. Immediate: every point of $\mathcal{H}$ lies in some such compact K. □

Remark 7.8 (Where this leaves the conjecture). Sufficiency now reads: $\widehat{X} \setminus X$ always ([2, Thm. 3.1]); defective horoballs always ([2, Thm. 4.1]); split horoballs under $\eta > L_\infty/2$, hence unconditionally when $L_\infty = 0$ (Theorem 7.5, Corollary 7.6). Two gaps remain. On the sufficiency side: the regime $0 < 2\eta \le L_\infty$, where the horoball is split only at scales below half the asymptotic separation of the contact rays (possible when those rays bound a flat half-strip); there neither Theorem 7.5 nor defectiveness applies. On the necessity side: the branches of §5 deliver maximal free horoballs with at least one attained contact (drift: subsequential limits of feet; ray escape: the inflation foot), and for reconstructible p these must be upgraded to *split-or-defective* (Question 12.1). By Proposition 3.1 the upgrade cannot come from a single inflation: the second contact cluster must be injected by equidistant motion of centres, for which the pincer of Theorem 7.5 is the prototype.

8 Progress on the split upgrade

Throughout this section $p \in X^c \setminus \widehat{X}$. Normalise all Busemann functions at a basepoint ($b_\xi(o) = 0$); then $(\xi, y) \mapsto b_\xi(y)$ is continuous on $\partial_\infty M \times M$ [4, II.8], so $c_\xi := \inf_X b_\xi \in [-\infty, \infty)$ is upper semicontinuous in ξ and the *depth function*

$$D(\xi) \ := \ c_\xi - b_\xi(p)$$

is upper semicontinuous on the compact $\partial_\infty M$; it is independent of the normalisation, positive somewhere precisely because $p \notin \widehat{X}$, and attains its maximum $\delta^\star := \max_\xi D(\xi) > 0$ at some $\xi^\star$.

Theorem 8.1 (Drift or asymptotic tangency)**.** *Let $p \in X^c \setminus \widehat{X}$, let $\gamma^\star$ be a ray asymptotic to a depth-maximising $\xi^\star$, and write $b^\star := b_{\gamma^\star}$, $c^\star := c_{\gamma^\star}$. Then either $T_{\delta^\star/4}(p) = \infty$ (*drift*), or for all sufficiently large t the inflation from $\gamma^\star(t)$ through any foot escapes, and in that case:*

- *(a) $d(\gamma^\star(t)) = t + c^\star + o(1)$ (asymptotically exact free radii along the deepest axis);*
- *(b) every choice of feet $x_t \in m(\gamma^\star(t))$ satisfies $b^\star(x_t) \to c^\star$ (asymptotic contacts of the deepest horoball);*
- *(c) every subsequential limit ξ' of the escape directions is itself depth-maximising: $D(\xi') = \delta^\star$.*

Proof. For $x \in X$ and every t, the non-increasing limit defining $b^\star$ gives $d(x, \gamma^\star(t)) - t \ge b^\star(x) \ge c^\star$, so $d(\gamma^\star(t)) \ge t + c^\star$; set $\Delta_t := d(\gamma^\star(t)) - t - c^\star \ge 0$ and $\varepsilon(t) := d(p, \gamma^\star(t)) - t - b^\star(p) \downarrow 0$. The margin of p in the maximal free ball at $\gamma^\star(t)$ is

$$\mu_t \ = \ d(\gamma^\star(t)) - d(p, \gamma^\star(t)) \ = \ \Delta_t + \delta^\star - \varepsilon(t) \ > \ 0 \quad \text{for large } t.$$

If the inflations from $\gamma^\star(t)$ are cut for an unbounded set of t, Lemma 2.2 (with $\varepsilon = \mu_t/4$) produces $a'_t \in M_X$ with $d(a'_t) \ge d(\gamma^\star(t)) - \mu_t/4 \to \infty$ and $\mu(a'_t, p) \ge \mu_t/2 \ge (\delta^\star - \varepsilon(t))/2 \ge \delta^\star/4$ eventually: drift. Otherwise the inflation from $\gamma^\star(t)$ escapes for all large t; Theorem 3.3 places p at depth $\ge \mu_t$ in the maximal free horoball of the escape direction ξ_t, whence

$$\Delta_t + \delta^\star - \varepsilon(t) \ = \ \mu_t \ \le \ D(\xi_t) \ \le \ \delta^\star,$$

so $\Delta_t \le \varepsilon(t) \to 0$, proving (a). For (b), $c^\star \le b^\star(x_t) \le d(x_t, \gamma^\star(t)) - t = c^\star + \Delta_t$. For (c), $D(\xi_t) \ge \mu_t \to \delta^\star$ and upper semicontinuity give $D(\xi') \ge \delta^\star$ for any subsequential limit, while $\le \delta^\star$ by maximality. □

Remark 8.2. In the non-drift case the deepest horoball is asymptotically tangent along its own axis: the obstruction to drift is *exactly* an asymptotic-contact regime. The next example shows that this regime is, at least in the model case, the territory of defectiveness — the hull does *not* collapse against asymptotic strands.

Example 8.3 (Asymptotic strand: defective but not split). In the upper half-plane model of $\mathbb{H}^2$ let

$$X \;:=\; \{x_1\} \cup \{(u,\, 1-e^u) : u \le -1\}, \qquad x_1 := (0,1),$$

a contact point together with a strand asymptotic to the horosphere $\{y = 1\}$ from below. For the upward ray γ ($b_\gamma = -\log y$, $c_\gamma = 0$), the maximal free horoball $\mathcal{H}_\gamma = \{y > 1\}$ has the single attained contact x_1, so it is not split (a second cluster meeting S_γ does not exist). It *is* defective: $w := (-10, 1)$ is a trapped witness. Indeed, a free horoball containing w centred at ∞ is some $\{y > c\}$ with $c < 1$, which meets the strand (heights $\to 1$). A free horoball centred at $\xi \in \mathbb{R}$ is a Euclidean disk tangent at $(\xi, 0)$ of radius ρ; containing w forces $(\xi + 10)^2 + 1 < 2\rho$, freeness against x_1 forces $2\rho \le \xi^2 + 1$ (so $\xi < -5$), and then freeness against the strand point $(\xi, 1 - e^\xi)$ forces $2\rho \le 1 - e^\xi < 1$ — contradicting $2\rho > (\xi + 10)^2 + 1 \ge 1$. Hence w lies in no free horoball: $w \in (S_\gamma \cap \widehat{X}) \setminus X$, and $\mathcal{H}_\gamma \subseteq \mathcal{R}_X$ by [2, Thm. 4.1]. Defectiveness thus covers the asymptotic-contact regime in which splitness is unavailable, complementing Example 7.1.

Proposition 8.4 (Split upgrade on $\mathbb{H}^2$: non-degenerate drift)**.** *Let $M = \mathbb{H}^2$ and let p be in the drift case, with medial balls $B_k := B(a_k, r_k) \ni p$ (margins $\ge \delta$, $r_k \to \infty$) as in Theorem 4.2, whose feet pairs $q_k \ne q_k'$ subconverge to* distinct *contacts $x^\star \ne x^{\star\prime}$ of the limit horoball $\mathcal{H}^\star$. Then $\mathcal{H}^\star$ is split; consequently (Corollary 7.6) $\mathcal{H}^\star \subseteq \mathcal{R}_X$, and p lies in a split maximal free horoball at depth $\ge \delta$.*

Proof. Work in the upper half-plane model, normalised so that the limit direction is $\xi = \infty$, $b(x, y) = -\log y$, and the limit level s_∞ is the horosphere $\{y = Y\}$; then $X \subseteq \{y \le Y\}$ (Theorem 4.2(a)), the contacts are $x^\star = (u_1, Y)$, $x^{\star\prime} = (u_2, Y)$ with $G := u_2 - u_1 > 0$, and $b(x^\star) = s_\infty = c_\xi$, attained. Hyperbolic balls are Euclidean disks, so each B_k is a Euclidean disk; by the exhaustion property of Theorem 4.2(b) the B_k eventually contain Euclidean disks of any prescribed radius inside $\{y > Y\}$, hence their Euclidean radii satisfy $R_k \to \infty$.

Clearance. The boundary circle of B_k passes through $q_k \to (u_1, Y)$ and $q_k' \to (u_2, Y)$, whose heights are $\le Y$ (they lie in X). Let ℓ_k be the chord through them; $\ell_k \le Y$ on the segment, and the lower boundary arc lies below ℓ_k, with sagitta at horizontal distance $\ge G/4$ from both feet at least $2\beta_k$, where $\beta_k := G^2/(128 R_k) > 0$ (parabolic approximation of a circle of radius $R_k \gg G$; the feet positions converge to u_1, u_2, so for large k the middle half $I := [u_1 + G/4,\, u_2 - G/4]$ of the positions lies strictly between them). Consequently, for large k, every point with position in I and height in $[Y - \beta_k,\, Y]$ lies strictly between the lower arc and the upper arc, i.e. in the open disk B_k; freeness forbids X there.

Partition. In b-levels, heights $[Y - \beta_k, Y]$ are exactly $\{s_\infty \le b \le s_\infty + \eta_k\}$ with $\eta_k := -\log(1 - \beta_k/Y) > 0$, and points with $b \le s_\infty + \eta_k$ have height $\ge Y - \beta_k$. Hence $X \cap \{b \le s_\infty + \eta_k\}$ contains no point with position in I, and splits as $F_1 \cup F_2$ by position ($\le u_1 + G/4$ versus $\ge u_2 - G/4$). Both are closed, contain the contacts $x^\star$, $x^{\star\prime}$ respectively, and have positive hyperbolic separation: any path joining them either stays below height H at horizontal cost $\ge (G/2)/H$, or climbs to height H at cost $\ge 2\log(H/Y)$, so $\operatorname{dist}(F_1, F_2) \ge \min(G/(2Y),\, 2\log\frac{G}{4Y} + 2) > 0$. Thus $\mathcal{H}^\star$ is split at scale η_k; since $L_\infty = 0$ on $\mathbb{H}^2$, Theorem 7.5 applies at every scale. □

Remark 8.5 (The remaining core). After Theorem 8.1 and Proposition 8.4, what remains of Question 12.1 is exactly:

(i) *degenerate drift*: feet pairs merging ($d(q_k, q_k') \to 0$) or escaping to infinity — the clearance scale β_k and the separation both degenerate with G;

(ii) *the non-drift case*: upgrading the asymptotic tangency of Theorem 8.1 to defectiveness, for which Example 8.3 is the model;

(iii) *dimension* ≥ 3: the cleared region is a disk-shaped shadow in the horosphere $\cong \mathbb{R}^{n-1}$, whose complement is connected — contact chains may detour around it. This is not a mere proof-technique obstruction: §9 shows that for $n \geq 3$ the split-or-defective conjecture is *false*, and that splitness is a two-dimensional artefact. The remaining cases (i), (ii) are therefore moot for Conjecture 1.5 as stated; they survive only relative to the corrected notion of §9.

9 The conjecture fails in dimension three

The split refinement was forced by the collapse of the horoball hull (Remark 1.4); the two-point set in $\mathbb{H}^2$ recovered reconstruction through bitangency, two *separated* contacts. We now show that in $\mathbb{H}^3$ a single connected contact set already defeats Conjecture 1.5: splitness is a coincidence of dimension two.

Theorem 9.1 (A counterexample in $\mathbb{H}^3$)**.** *Work in the upper half-space model* $\mathbb{H}^3 = \{(x, y, z) : z > 0\}$*,* $\cosh d(p, q) = 1 + |p - q|^2/(2z_p z_q)$*, and let*

$$X \;=\; \{(\cos\theta, \sin\theta, 1) : \theta \in [0, 2\pi)\},$$

the unit circle on the horosphere $\{z = 1\}$*. Write* $R = \sqrt{x^2 + y^2}$ *and* $\mathcal{H}_0 = \{R^2 + (z - 1)^2 < 1\}$ *(the open Euclidean ball of centre* $(0, 0, 1)$ *and radius* 1*, a free horoball centred at the ideal point* 0*). Then*

(a) $\widehat{X} = X$*: the horoball hull collapses to the circle;*

(b) $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0$*, a non-empty open set; it is exactly the union of the two maximal free horoballs whose horospheres contain all of* X*;*

(c) every maximal free horoball meets X *in a connected set (either a single point or the whole circle); hence none is split, and by* (a) *none is defective;*

(d) consequently the right-hand side of Conjecture 1.5 is empty while $\mathcal{R}_X \neq \emptyset$*. The conjecture is false for* $n = 3$*.*

Proof. The medial axis is the z-axis: for $a = (0, 0, h)$, $h > 0$, $\cosh d(a, (\cos\theta, \sin\theta, 1)) = 1 + \frac{1+(h-1)^2}{2h} = \frac{h^2+2}{2h}$ independently of θ, so all of X is equidistant and a is medial; any point off the axis has, by strict monotonicity of $\cosh d$ in $-2R\cos\theta$, a unique nearest point, so is not medial.

(b) A point $p = (x, y, z)$ lies in the medial ball $B(a, d(a))$ iff $\cosh d(p, a) < \cosh d(a)$, i.e. $1 + \frac{R^2+(z-h)^2}{2zh} < \frac{h^2+2}{2h}$. Clearing denominators ($2zh > 0$) and cancelling $-2zh$,

$$R^2 + z^2 - 2z + (1 - z)\, h^2 \;<\; 0. \tag{$*$}$$

If $z > 1$ the coefficient $(1 - z)$ is negative, so ($*$) holds for h large: $\{z > 1\} \subseteq \mathcal{R}_X$. If $z = 1$, ($*$) reads $R^2 < 1$. If $z < 1$, the left side is increasing in h^2, with infimum $R^2 + z^2 - 2z$ as $h \to 0^+$, so $(*)$ is solvable iff $R^2 + (z-1)^2 < 1$. Since off-axis points are not medial, these axis balls give all of $\mathcal{R}_X$; their union is $\{z > 1\} \cup \mathcal{H}_0$. The horospheres $\{z = 1\}$ (ideal point ∞) and $\partial\mathcal{H}_0$ (ideal point 0) both contain the entire circle.

(a) Each $q = (x_0, y_0, z_0) \notin X$ lies in a free open horoball: if $z_0 > 1$, in $\{z > 1\}$; if $R_0^2 + (z_0 - 1)^2 < 1$, in $\mathcal{H}_0$; otherwise $z_0 \le 1$, $R_0 > 0$, and the horoball centred at the ideal point $(x_0, y_0, 0)$ — the Euclidean ball of centre (x_0, y_0, ρ) and radius ρ — is free iff $\rho \le \frac{1}{2}((R_0 - 1)^2 + 1)$ and contains q iff $\rho > z_0/2$; these are compatible because $z_0 - 1 < (R_0 - 1)^2$ (strict, as $q \notin X$). Hence $M \setminus X$ is covered by free horoballs and $\widehat{X} = X$.

(c) A maximal free horoball is centred at an ideal point ξ. For $\xi = \infty$ it is $\{z > 1\}$, contact set the whole circle. For finite ξ of Euclidean norm σ it is the Euclidean ball of centre (ξ, ρ^*), radius $\rho^* = \frac{1}{2}((\sigma-1)^2+1)$, and the squared Euclidean distance from its centre to $(\cos\theta, \sin\theta, 1)$ equals $(\rho^* - 1)^2 + \sigma^2 - 2\sigma\cos\theta' + 1$ (θ' measured from ξ), minimised at the unique $\theta' = 0$ when $\sigma > 0$ — one contact — and constant when $\sigma = 0$ — the whole circle ($\mathcal{H}_0$). In every case the contact set is connected, so cannot be partitioned into two non-empty closed pieces at positive distance: no maximal free horoball is split. By *(a)*, $\widehat{X} \setminus X = \varnothing$, so none is defective either.

(d) The right-hand side of Conjecture 1.5 is $(\widehat{X} \setminus X) \cup \bigcup\{\text{split or defective}\} = \varnothing$, whereas $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0 \neq \varnothing$. □

Remark 9.2 (Splitness is a two-dimensional artefact). A horosphere S_γ of a Hadamard manifold is intrinsically flat, $S_\gamma \cong \mathbb{E}^{n-1}$. The reconstruction in Theorem 9.1 is driven by the contact set $K = S_\gamma \cap X$ failing to be *intrinsically convex* in S_γ: the circle's flat convex hull is the disk, and the reconstructed points sit over the disk. In dimension $n = 2$ the horosphere is a line, $S_\gamma \cong \mathbb{E}^1$, where "convex hull strictly larger than K" is the same as "K has a gap", i.e. K disconnects — which is precisely splitness. For $n - 1 \ge 2$ a connected set (the circle) can have strictly larger convex hull, so splitness (a connectivity property) is strictly weaker than the operative condition and misses the circle entirely. This is the exact mechanism by which Conjecture 1.5 fails.

Definition 9.3 (Horospherical defect)**.** *For a maximal free horoball $\mathcal{H}_\gamma$ let $K_\gamma := S_\gamma \cap X$ be its contact set and let* $\mathrm{hconv}_\gamma(K_\gamma)$ *be the closed convex hull of K_γ taken in the intrinsic flat metric of the horosphere $S_\gamma \cong \mathbb{E}^{n-1}$. Call $\mathcal{H}_\gamma$* horospherically defective *if* $\mathrm{hconv}_\gamma(K_\gamma) \neq K_\gamma$.

Proposition 9.4 (The corrected notion passes the tests)**.**

- *(a)* (Circle.) *In Theorem 9.1 the horospherically defective maximal free horoballs are exactly $\{z > 1\}$ and $\mathcal{H}_0$ (contact set the whole circle, flat hull the disk); their union is $\mathcal{R}_X$. The single-contact horoballs are not horospherically defective.*
- *(b)* (Euclidean reduction.) *For $M = \mathbb{R}^n$ and compact X, $\mathcal{H} = L^+$ is horospherically defective iff the slice $L \cap X$ is non-convex iff L is defective in the sense of [1] (Proposition 6.1(c)). Thus Definition 9.3 reduces to the hypothesis of Theorem 6.*
- *(c)* (Planar reduction.) *For $M = \mathbb{H}^2$ a horoball whose contact set is closed equals / refines splitness: a contact set $K \subset S \cong \mathbb{E}^1$ has* $\mathrm{hconv}(K) \neq K$ *iff K has a gap, the on-the-horosphere shadow of the split condition (Definition 7.2).*

Proof. (a) Computed in the proof of Theorem 9.1(c): only $\xi \in \{0, \infty\}$ give multi-point (whole-circle) contact, with flat hull the disk $\neq$ circle; finite $\xi \neq 0$ give a single contact, flat hull a point. The union of the two is $\{z > 1\} \cup \mathcal{H}_0 = \mathcal{R}_X$ by (b) of that theorem. (b) For compact X the face $\overline{\mathrm{conv}}\, X \cap L = \mathrm{conv}(L \cap X)$, so the ambient defect witness $w \in (\overline{\mathrm{conv}}\, X \cap L) \setminus X$ exists iff

$\operatorname{conv}(L \cap X) \neq L \cap X$; and on the hyperplane L the intrinsic and ambient convex hulls coincide. (c) On a line the closed convex hull of K is the smallest interval containing it, which differs from K exactly when K omits an interior point, i.e. has a gap. $\square$

Conjecture 9.5 (Corrected necessity-and-sufficiency)**.** *Let M be Hadamard and X closed. Then*

$$\mathcal{R}_X \;=\; (\widehat{X} \setminus X) \;\cup\; \bigcup \Big\{ \mathcal{H}_\gamma \;:\; \mathcal{H}_\gamma \textit{ maximal free, horospherically defective} \Big\}.$$

Three caveats keep this honest. First, the strand (Example 8.3) shows that contacts can be *asymptotic* — X approaching S_γ without meeting it — so that K_γ undercounts; there $\mathcal{H}_\gamma$ is defective in the ambient sense but not horospherically defective, so the corrected union must be read as horospherically defective *or* defective, and the genuinely correct object is presumably the flat convex hull of the *asymptotic* contact set $\overline{\{x \in X : b_\gamma(x) \to c_\gamma\}}$ in a suitable horospherical compactification (Question 12.2). Second, for a horospherically defective $\mathcal{H}_\gamma$ it is the points lying *over* $\operatorname{hconv}_\gamma(K_\gamma) \setminus X$ that should be reconstructed; whether this is the whole horoball (as for the circle and in $\mathbb{R}^n$) or a proper subregion in general needs the inflation analysis of §2 re-run with horospherical feet. Third, sufficiency ($\supseteq$) is now itself open in dimension ≥ 3: the split sufficiency Theorem 7.5 must be upgraded to a horospherical-hull statement, the natural vehicle being a *wall* of medial centres swept over a triangulation of $\operatorname{hconv}_\gamma(K_\gamma)$ rather than the single segment J_t of the planar pincer.

10 Sufficiency for horospherically defective horoballs

We now prove the $\supseteq$ half of Conjecture 9.5 in real hyperbolic space $\mathbb{H}^n$ — where the horospheres are genuinely flat, $S_\gamma \cong \mathbb{E}^{n-1}$, so Definition 9.3 is unambiguous. The mechanism is a clean duality with a classical theorem.

Theorem 10.1 (Motzkin [7])**.** *A non-empty closed set $C \subseteq \mathbb{E}^N$ is convex if and only if it is* Chebyshev*: every point of $\mathbb{E}^N$ has a unique nearest point in C. Equivalently, C is non-convex iff some point has at least two nearest points in C.*

We use the upper half-space model $\mathbb{H}^n = \{(\xi, z) : \xi \in \mathbb{R}^{n-1}, z > 0\}$ with $\cosh d(p,q) = 1 + |p-q|^2/(2z_p z_q)$. Isometries act transitively on horoballs and restrict to Euclidean similarities of the flat horosphere (preserving convexity), so for any maximal free horoball it costs nothing to assume

$$\mathcal{H} = \{z > 1\}, \quad S = \{z = 1\} \cong \mathbb{R}^{n-1}, \quad X \subseteq \{z \leq 1\}, \quad K = X \cap \{z = 1\}.$$

For $a = (\alpha, t)$ and a contact $k = (\kappa, 1)$, $\cosh d(a,k) = 1 + \frac{|\alpha-\kappa|^2 + (t-1)^2}{2t}$ is strictly increasing in the *Euclidean* position distance $|\alpha - \kappa|$; so among contacts the feet of a are the Euclidean-nearest elements of K, and the medial directions are the non-Chebyshev points of K.

Theorem 10.2 (Reconstruction on a horosphere)**.** *Let X lie on a horosphere of $\mathbb{H}^n$ (so $K = X$). Then:*

(a) if X is not convex in $S \cong \mathbb{E}^{n-1}$, the entire horoball is reconstructed, $\mathcal{H} \subseteq \mathcal{R}_X$;

(b) if X is convex, then $M_X = \varnothing$ and $\mathcal{R}_X = \varnothing$.

Proof. Every point of X has height 1, so for any $a = (\alpha, t)$ the feet $m(a)$ are exactly the Euclidean-nearest positions of $K = X$ to α, for every $t > 0$.

(b) If X is convex, Motzkin gives a unique nearest position for each α, so every a has a single foot: $M_X = \varnothing$.

(a) If X is non-convex, Motzkin gives α^* with two distinct nearest contacts k_1, k_2 at position distance $\rho := \rho_K(\alpha^*) > 0$. Then $a_t := (\alpha^*, t)$ has $k_1, k_2 \in m(a_t)$ for every $t > 0$: it is medial, with $\cosh d(a_t) = 1 + \frac{\rho^2+(t-1)^2}{2t}$. For $p = (\pi, z) \in \mathcal{H}$ (so $z > 1$), $p \in B(a_t, d(a_t))$ iff $\cosh d(p, a_t) < \cosh d(a_t)$, which after clearing denominators and cancelling $-2zt$ reads

$$(1-z)\, t^2 \;+\; \big(|\pi - \alpha^*|^2 + z^2 - z\rho^2 - z\big) \;<\; 0.$$

Since $z > 1$ the leading coefficient $1 - z$ is negative, so the inequality holds for all t large enough: $p \in \mathcal{R}_X$. As $p \in \mathcal{H}$ was arbitrary, $\mathcal{H} \subseteq \mathcal{R}_X$. □

Corollary 10.3 (The circle, completed)**.** *For the unit circle of Theorem 9.1, both maximal free horoballs with whole-circle contact — $\{z > 1\}$ and $\mathcal{H}_0$ — satisfy $\mathcal{H} \subseteq \mathcal{R}_X$, because the circle is non-convex on each of their (flat) horospheres. Hence $\mathcal{R}_X = \{z > 1\} \cup \mathcal{H}_0$ equals the union of the horospherically defective horoballs, and Conjecture 9.5 holds, as an equality, for this set. The example that refuted Conjecture 1.5 thus confirms its repair.*

When X leaves the horosphere, the only obstruction is interference: sub-horosphere mass could beat a contact for the chosen medial direction. A uniform separation of X from the horosphere removes it.

Theorem 10.4 (Reconstruction, isolated contacts)**.** *Let $\mathcal{H} = \{z > 1\}$ be maximal free in $\mathbb{H}^n$ with contact set K, and suppose the contacts are* isolated from the bulk*: $X \cap \{z > 1 - \nu\} = K$ for some $\nu > 0$. If K is non-convex in $S \cong \mathbb{E}^{n-1}$ (i.e. $\mathcal{H}$ is horospherically defective), then $\mathcal{H} \subseteq \mathcal{R}_X$.*

Proof. Take α^* and k_1, k_2, ρ as before (Motzkin on the closed set K). For $x = (\chi, s) \in X$ with $s \le 1 - \nu$,

$$\cosh d(a_t, x) - \cosh d(a_t, k_1) \;\ge\; \frac{(t-s)^2}{2ts} - \frac{\rho^2 + (t-1)^2}{2t} = \frac{(1-s)(t^2 - s) - s\rho^2}{2ts},$$

and $(1-s)(t^2-s) - s\rho^2 \ge \nu(t^2-1) - \rho^2 > 0$ once $t > \sqrt{1 + \rho^2/\nu}$ — a bound independent of x. So beyond this t no bulk point beats the nearest contacts, $m(a_t) \subseteq K$ consists of the (two or more) Euclidean-nearest contacts, and a_t is medial. The reconstruction inequality of Theorem 10.2(a) is unchanged (it used only the contacts, at height 1), so $\mathcal{H} \subseteq \mathcal{R}_X$. □

Remark 10.5 (What is proved, and what remains). Theorems 10.2–10.4 prove sufficiency for every horospherically defective horoball in $\mathbb{H}^n$ whose contacts are attained and isolated from the bulk; in particular the corrected conjecture's $\supseteq$ holds, and Corollary 10.3 closes the circle counterexample. Three frontiers remain. *(i) Tangential bulk.* If X approaches the horosphere tangentially at a contact (mass $(\chi_j, 1 - \eta_j)$ with $\eta_j = o(|\chi_j - \kappa|)$), a fixed medial direction can be shadowed; one expects reconstruction to survive through a *moving* medial direction, but the clean Motzkin argument no longer applies verbatim. *(ii) Curved horospheres.* Outside $\mathbb{H}^n$ the horosphere need not be flat, and even "intrinsically convex" must be reread (the induced metric is a Hadamard metric only under curvature pinching); Motzkin's theorem must be replaced by its Chebyshev analogue in the relevant Alexandrov class. *(iii) Necessity.* The reverse inclusion is the horospherical form of the upgrade (Question 12.1): a reconstructible $p \notin \widehat{X}$ sits in a medial ball whose two feet, followed along the structure trichotomy of §5, must be shown to limit onto a *non-convex* contact set. By Motzkin this is equivalent to the limit contacts failing to be Chebyshev — exactly the multiplicity that the two feet $q \neq q'$ encode, provided they neither merge nor are filled in by X in the limit. This reduces necessity to a quantitative non-merging estimate, the higher-dimensional descendant of the planar Proposition 8.4.

11 Necessity: the Motzkin reduction and the non-merging estimate

We turn to $\subseteq$ in Conjecture 9.5. The same duality controls it: a reconstructible point outside the hull is reconstructed by medial centres whose feet, followed along the drift of §5, must ascend to the horosphere and land on a *non-Chebyshev* — hence by Motzkin non-convex — contact set. The one thing that could go wrong is that the two feet *merge* in the limit; the estimate below shows they cannot, except by collapsing all the way onto a single contact, which is a measure-zero event we isolate precisely.

Throughout, $\mathcal{H} = \{z > 1\}$ is maximal free in $\mathbb{H}^n$, $X \subseteq \{z \leq 1\}$, $K = X \cap \{z = 1\} \neq \varnothing$, and for $a = (\alpha, t)$, $x = (\chi, s)$ recall $\cosh d(a, x) = 1 + \frac{|\alpha-\chi|^2+(t-s)^2}{2ts}$, monotone in $|\alpha - \chi|$ at fixed heights.

Proposition 11.1 (Contact feet force a defect)**.** *If some medial centre a reconstructing p has two feet $q \neq q'$ both lying in K, then K is non-convex and $\mathcal{H}$ is horospherically defective. No further hypotheses.*

Proof. q, q' are nearest points of X to $a = (\alpha, t)$, both in $K \subseteq \{z = 1\}$. By monotonicity at height 1, their positions κ, κ' are Euclidean-nearest to α in K, at a common distance; since $\kappa \neq \kappa'$, the point α has two nearest points in K. So K is not Chebyshev, hence by Theorem 10.1 not convex. $\square$

Lemma 11.2 (Ascent)**.** *Let $a_k = (\alpha_k, t_k)$ be medial with $t_k \to \infty$ and $\alpha_k \to \alpha_\infty$, and let $q_k = (\beta_k, s_k) \in m(a_k)$ have bounded positions β_k. Then $1 - s_k = O(t_k^{-2})$; in particular every subsequential limit of q_k lies in K, and $h_k := (1 - s_k)t_k^2$ is bounded.*

Proof. Fix a contact $k_0 = (\kappa_0, 1) \in K$. As q_k is a nearest point, $\cosh d(a_k, q_k) \leq \cosh d(a_k, k_0)$, i.e.

$$\frac{|\alpha_k - \beta_k|^2 + (t_k - s_k)^2}{s_k} \;\leq\; |\alpha_k - \kappa_0|^2 + (t_k - 1)^2.$$

Dropping the non-negative position term and using $\frac{(t_k-s_k)^2}{s_k} - (t_k - 1)^2 = \frac{(1-s_k)(t_k^2-s_k)}{s_k}$ gives $\frac{(1-s_k)(t_k^2-s_k)}{s_k} \leq |\alpha_k - \kappa_0|^2 \leq C$, whence $(1 - s_k)t_k^2 \leq C'$ for large k. Closedness of X then places limits of q_k in $X \cap \{z = 1\} = K$. $\square$

Theorem 11.3 (Non-merging estimate)**.** *Let $a_k = (\alpha_k, t_k)$ be medial centres of a drift reconstructing $p \in \mathcal{H}$ ($t_k \to \infty$, $\alpha_k \to \alpha_\infty$, margins bounded below), with distinct feet $q_k \neq q'_k$ of bounded position. Passing to a subsequence, let $q_k \to q_* = (\kappa_*, 1)$ and $q'_k \to q'_* = (\kappa'_*, 1)$ in K (Lemma 11.2). Then:*

(a) the heights ascend sharply, $h_* = \lim(1-s_k)t_k^2 = 0$ *and likewise for* q'_k*; both* κ_*, κ'_* *are Euclidean-nearest to* α_∞ *in* K*;*

(b) (estimate) *either* $\kappa_* = \kappa'_*$ *(*collapse*), or* $\kappa_* \neq \kappa'_*$ *and then*

$$d(q_k, q'_k) \;\longrightarrow\; \operatorname{arccosh}\Big(1 + \tfrac{1}{2}|\kappa_* - \kappa'_*|^2\Big) \;>\; 0;$$

there is no intermediate decay rate;

(c) in the non-collapse case K *has two distinct nearest points to* α_∞*, so by Motzkin it is non-convex:* $\mathcal{H}$ *is horospherically defective and contains* p*. Necessity holds.*

Proof. Write $h_k = (1-s_k)t_k^2$, $h'_k = (1-s'_k)t_k^2$, bounded by Lemma 11.2; pass to limits $h_*, h'_* \geq 0$. For any contact $(\kappa, 1)$ the nearest-point inequality $\cosh d(a_k, q_k) \leq \cosh d(a_k, (\kappa, 1))$, multiplied by $2t_k$ and rearranged with $\frac{(t_k - s_k)^2}{s_k} - (t_k - 1)^2 = \frac{(1-s_k)(t_k^2 - s_k)}{s_k} \to h_*$, passes to the limit as

$$|\alpha_\infty - \kappa_*|^2 + h_* \;\leq\; |\alpha_\infty - \kappa|^2 \qquad (\forall\, \kappa \in K), \tag{†}$$

and symmetrically $|\alpha_\infty - \kappa'_*|^2 + h'_* \leq |\alpha_\infty - \kappa|^2$. Putting $\kappa = \kappa'_*$ in the first and $\kappa = \kappa_*$ in the second and adding yields $h_* + h'_* \leq 0$, so $h_* = h'_* = 0$: this proves (a), and (†) becomes $|\alpha_\infty - \kappa_*| = |\alpha_\infty - \kappa'_*| = \min_{\kappa \in K}|\alpha_\infty - \kappa|$.

For (b), $d(q_k, q'_k) \to d(q_*, q'_*)$, and for two points at height 1, $\cosh d(q_*, q'_*) = 1 + \frac{1}{2}|\kappa_* - \kappa'_*|^2$. If $\kappa_* \neq \kappa'_*$ this is > 1, the stated bound; the absence of intermediate rates is the dichotomy "equal or not". For (c), distinct equidistant nearest points show K non-Chebyshev; Motzkin gives non-convexity, and p lies in $\mathcal{H} = \{z > 1\}$ by construction. $\square$

Remark 11.4 (The collapse residue). The estimate leaves exactly one way for necessity to be threatened: *collapse*, $q_k, q'_k \to$ a single contact κ_*. Three remarks bound its reach. First, collapse cannot occur over a *smooth* contact: if near κ_* the set X is a graph $z = 1 - \phi$ with $\phi \geq 0$ strictly convex (a rounded touching), the high-centre feet minimise $|\alpha - \chi|^2 + t^2\phi(\chi)$ which is strictly convex, giving a *unique* foot — a_k is then not medial. Collapse therefore forces a non-smooth (cusp or concave-corner) contact, where K is already non-convex, so Proposition 11.1/Motzkin still apply provided the feet are eventually *in* K. Second, the genuinely residual case is collapse with feet strictly below the horosphere hugging an isolated contact; there the reconstructed points lie within $o(1)$ of κ_*, and whether they fall in $\widehat{X}$ (first term, nothing to prove) or expose an ambient defect is exactly the local-hull content of the asymptotic-contact Question 12.2. Third, when contacts escape to infinity (β_k unbounded) the limit is asymptotic and again governed by Question 12.2 (the strand, Example 8.3). Thus: *necessity is proved in* $\mathbb{H}^n$ *for every drift whose feet stay bounded and do not collapse onto one contact; the remainder reduces, with no loss, to the single asymptotic/local-hull question already isolated as open.*

12 Summary and open problems

Result	Hypothesis	Status
$\widehat{X} \setminus X \subseteq \mathcal{R}_X$	complete	proved [2, Thm. 3.1]
defective $\mathcal{H} \subseteq \mathcal{R}_X$	Hadamard	proved [2, Thm. 4.1]
No escape from medial centres (rigidity)	Hadamard	proved (Prop. 3.1)
Drift $\Rightarrow$ maximal $\mathcal{H} \ni p$, contacts captured	Hadamard	proved (Thm. 4.2)
Trichotomy hull/drift/ray-escape, no residual case	Hadamard	proved (Thm. 5.1)
$\mathcal{R}_X \setminus \widehat{X} \subseteq$ drift	$\mathbb{R}^n$	proved (Prop. 5.3)
Drift or asymptotic tangency	Hadamard	proved (Thm. 8.1)
Split upgrade, non-degenerate drift	$\mathbb{H}^2$	proved (Prop. 8.4)
Defective $\setminus$ split $\neq \emptyset$ (strand)	$\mathbb{H}^2$	example (Ex. 8.3)
Split sufficiency, $\eta > L_\infty/2$	$\mathbb{H}^2$/pinched	proved (Thm. 7.5, Cor. 7.6)
Split-or-defective conjecture	$n \geq 3$	**false** (Thm. 9.1)
Splitness is a 2-dim artefact	—	Rem. 9.2
Horospherical defect: circle test	$\mathbb{H}^3$	proved (Prop. 9.4a)
Horospherical defect = defective	$\mathbb{R}^n$	proved (Prop. 9.4b)
Horospherical defect $\approx$ split	$\mathbb{H}^2$	proved (Prop. 9.4c)
Corrected conjecture	Hadamard	open (Conj. 9.5)
Horosph. defective $\mathcal{H} \subseteq \mathcal{R}_X$, X on horosphere	$\mathbb{H}^n$	proved (Thm. 10.2)
X convex on horosphere $\Rightarrow \mathcal{R}_X = \varnothing$	$\mathbb{H}^n$	proved (Thm. 10.2)
Horosph. defective $\mathcal{H} \subseteq \mathcal{R}_X$, isolated contacts	$\mathbb{H}^n$	proved (Thm. 10.4)
Circle: $\mathcal{R}_X$ = union of horosph. defective	$\mathbb{H}^3$	proved (Cor. 10.3)
Sufficiency, tangential bulk / curved horosph.	Hadamard	open (Rem. 10.5)
Contact feet $\Rightarrow$ defect (Motzkin)	$\mathbb{H}^n$	proved (Prop. 11.1)
Ascent of bounded feet to K	$\mathbb{H}^n$	proved (Lem. 11.2)
Non-merging estimate $\Rightarrow$ necessity	$\mathbb{H}^n$	proved (Thm. 11.3)
Necessity, collapse / asymptotic residue	$\mathbb{H}^n$	open = Q. 12.2 (Rem. 11.4)

Question 12.1 (The necessity gap, horospherical form)**.** *Theorem 9.1 retires the split-or-defective form of the question; the live version is whether Conjecture 9.5 holds: for $p \in \mathcal{R}_X \setminus \widehat{X}$, must some maximal free horoball containing p be horospherically defective (Definition 9.3)? By Proposition 3.1 the upgrade cannot come from a single inflation: the medial witness of p injects its further contacts through equidistant motion of centres, as in the pincer of Theorem 7.5. The concrete target is an* equidistant-flow theorem*: starting from $a \in M_X$ and $p \in B(a, d(a))$, grow free balls with centres on the equidistant locus of the contact set, and show the family drifts to a horoball whose horospherical contact hull contains the shadow of p. For $n = 2$ this is the split upgrade of §8 (Remark 8.5 delimits its open core); for $n \geq 3$ the single segment J_t of the planar pincer must become a swept $(n-1)$-dimensional wall over* $\mathrm{hconv}_\gamma(K_\gamma)$.

Question 12.2. *Formulate the correct asymptotic notion of horospherical contact hull — the flat convex hull in $S_\gamma \cong \mathbb{E}^{n-1}$ of the* asymptotic *contact set $\overline{\{x \in X : b_\gamma(x) \to c_\gamma\}}$, in a horospherical compactification — so that Conjecture 9.5 both covers the asymptotic strand (Example 8.3, Proposition 6.1(b)), where contacts recede to the ideal boundary, and reduces to Definition 9.3 when contacts are attained. Then identify, for a horospherically defective horoball, the exact reconstructed subregion (the second caveat after Conjecture 9.5), and prove the resulting equality.*

Question 12.3. *Theorem 3.3 uses only global minimality of the extended foot-geodesic. On a complete, non-Hadamard manifold the statement survives whenever the inflation ray is globally minimizing; the flat cylinder (where bitangency without multiplicity is possible, $X = S^1 \times \{0\}$) shows that bitangency must then be read as "two minimizing geodesics". Determine the sharp generality.*

References

[1] A. Białożyt, *Sets reconstructible with medial axis*, preprint, AGH University of Kraków, 2024.

[2] *Sets reconstructible with medial axis: a generalisation to Riemannian manifolds*, companion analysis notes, June 2026 (file `medial_axis_riemannian_1.pdf`).

[3] *Formalization notes on Theorem 6 of [1]*, June 2026 (files `medial_formalization_notes.tex`, `medial_formalization_plan.md`); Lean 4 formalization on branch `medial-axis`, files `Mathlib/Analysis/Convex/MedialAxis*.lean`.

[4] M. Bridson and A. Haefliger, *Metric Spaces of Non-Positive Curvature*, Grundlehren 319, Springer, 1999.

[5] P. Albano, *On the cut locus of closed sets*, Nonlinear Anal. **125** (2015), 398–405.

[6] F.-E. Wolter, *Cut loci in bordered and unbordered Riemannian manifolds*, PhD thesis, TU Berlin, 1985.

[7] T. Motzkin, *Sur quelques propriétés caractéristiques des ensembles bornés non convexes*, Atti Accad. Naz. Lincei **21** (1935), 773–779. (Closed Chebyshev sets in $\mathbb{E}^N$ are convex.)